Supplementary Material A

Asymptotic theory of the adjusted ersatz discrepancy

to *An Adjusted Ersatz Discrepancy for Global Sensitivity Analysis: Theory and Benchmarks*

1 Overview

This supplement characterises the statistic proposed in the main text, the adjusted ersatz discrepancy, and its two building blocks. All results are stated for a single input X_i. Every proposition, theorem and corollary is proved. Where a proof uses a numerical verification of an explicit inequality in one or two real variables, it is marked computer-assisted and the verified inequality is stated. Values obtained by numerical optimisation, quadrature or simulation are labelled as such. Table 1 lists the results. The script `theory_checks.py` of the repository reproduces every numerical verification and every constant quoted below, and `null_distribution.py` reproduces Table 2. Constants defined by an equation in one variable (κ_λ, p_0, p_1, λ_c and the optimiser of Remark 7.3) are computed in multiple-precision arithmetic (30 to 40 significant digits). The computer-assisted inequalities are verified on explicit grids by monotone bracketing, and the verified margins ($6 \cdot 10^{-3}$ and more) exceed the rounding error of double precision by many orders of magnitude. The other values are rounded to the digits shown.

In words, the main conclusions are the following. After the rank transformation, the occupancy grid without fill equals the plug-in histogram estimator of the moment-independent measure δ_i of Borgonovo (2007). At the resolution used by the ersatz, about one point per cell, this estimator does not converge to δ_i. Its limit is e^{-1} plus a strictly convex f-divergence between the copula of (X_i, Y) and the independence copula, so that its value under independence is $e^{-1} \approx 0.368$. The Moore-neighbourhood fill changes the limit into a functional $\Psi_\lambda(C_i)$ that lies within $[\delta_i - 0.1401,\ \delta_i + 0.0471]$ for every dependence structure (default grid, $\lambda = 1$, with a computer-assisted upper bound) and whose minimum, $\kappa_1 = 0.0471$, is attained exactly under independence. Under the regularity assumption and simple random sampling, the adjusted ersatz converges almost surely to $\Psi_\lambda(C_i)$ when the output distribution is continuous, and in probability when it has atoms and ties are broken by order of appearance. The adjusted ersatz is therefore a fast, consistent estimator of a near-δ importance measure, but not of δ_i itself. Whenever $T_i = 0$ its limit is the value under independence, κ_λ, which is 0.0471 on the default grid and not zero. At finite N the statistic of such an input is random, with the null distribution of Section 9. The limit, and hence for large N the statistic (Corollary 4.5), ranks two inputs as δ does whenever their δ values differ by more than 0.18711. Numerically the sharp margin is 0.176: two inputs whose δ values differ by less than that can be ranked in either order. Theorem 4.1 is proved for simple random sampling. For an input independent of the output it extends to plain Latin hypercube sampling (Proposition 8.1), whereas for maximin Latin hypercubes and scrambled Sobol' sequences the evidence is simulation only (Sections 8 and 9). The phase transition in the grid exponent (Theorem 6.1) is established for $\alpha_g > 1/4$.

2 Setting, notation and statistics

Model and sampling. The inputs $\mathbf{X} = (X_1, \ldots, X_k)$ are mutually independent with continuous distribution functions and, without loss of generality, uniformly distributed on $[0, 1]$ (the

Result	Content	Status
Lemma 3.1	Ties in the output: order of appearance is equivalent to random tie-breaking under simple random sampling	proof
Proposition 3.2	Range of the statistics; the Moore fill can only decrease the statistic	proof
Proposition 3.3	The rank grid without fill is the histogram estimator of Borgonovo's δ	proof
Proposition 3.4	Finite-sample zero condition for a grid fixed in advance	proof
Theorem 4.1	Almost sure limits $\Phi_\lambda(C_i)$ and $\Psi_\lambda(C_i)$ under simple random sampling (in probability if the output has atoms and ties are broken by order of appearance), and exponential concentration	proof
Corollary 4.5	Consistency of the ranking and of a screening test for independence	proof
Proposition 5.1	$\Phi_\lambda - e^{-\lambda}$ is an f-divergence (an f-sensitivity index)	proof
Proposition 5.2	Bounds between Φ_λ and δ; the upper bound is sharp	proof
Proposition 5.4	Zero condition, its failure on fine grids, approximation of δ by Ψ_λ	proof; (iii) and (iv) computer-assisted; the threshold 0.7925 numerical
Proposition 5.6	The original S-ersatz depends on the output marginal, and tends to one for unbounded outputs	proof
Theorem 6.1	Phase transition in the grid exponent, for $\alpha_g > 1/4$	proof
Proposition 7.1	Relations with Sobol' indices; what a scatterplot shows	proof
Theorem 7.2	Rankings by T_i, S_i and every f-sensitivity index agree in single-index Gaussian models	proof
Remark 7.3	Rankings by Ψ_λ: a guarantee, its sharp margin and counterexamples	proof; sharp margin and Gaussian case numerical
Proposition 8.1	Latin hypercube sampling under independence	proof
Section 8	Maximin Latin hypercube and scrambled Sobol' designs	simulation
Section 9	Null distribution; meaning of the three thresholds of the main text	simulation
Section 10	Gaussian copulas: limits against simulation	numerical

Table 1: Results of this supplement.

estimators are applied to the probability-integral-transformed inputs, and an input with atoms would first have to be replaced by its distributional transform, defined below for the output). The output is $Y = f(\mathbf{X})$ with f measurable. A sample $(\mathbf{X}^{(j)}, Y_j)$, $j = 1, \dots, N$, $Y_j = f(\mathbf{X}^{(j)})$, is drawn by simple random sampling (SRS), that is, the $\mathbf{X}^{(j)}$ are independent and uniform on $[0,1]^k$. Almost sure statements refer to nested samples (the first N terms of one infinite sequence), and "almost everywhere" and "almost every" refer to Lebesgue measure on $[0,1]^2$. We fix an input i and write $U_j = X_i^{(j)}$.

Copula. Let F be the distribution function of Y and $\zeta \sim U(0,1)$ independent of $\mathbf{X}$. The distributional transform $V = F(Y-) + \zeta\,[F(Y) - F(Y-)]$ is uniform on $(0,1)$ and $F^{-1}(V) = Y$ almost surely (Rüschendorf, 2009, Proposition 2.1). The law C_i of (X_i, V) is therefore a copula of (X_i, Y) (Rüschendorf, 2009, Theorem 2.2); if F is continuous, $V = F(Y)$ and C_i is the unique copula of (X_i, Y) (Sklar, 1959; Nelsen, 2006). We write $C_i = C_i^{\mathrm{ac}} + C_i^{\mathrm{s}}$ for the Lebesgue decomposition, c_i for the density of the absolutely continuous part, $m_i = C_i^{\mathrm{s}}([0,1]^2)$ for the mass of the singular part, and Π for the independence copula. Note that $\int c_i \,\mathrm{d}u\,\mathrm{d}v = 1 - m_i$.

Assumption 2.1 (regularity)**.** There is a closed set $K \subset [0,1]^2$ with $\mathrm{Leb}(K) = 0$, containing the boundary of the unit square, such that $C_i^{\mathrm{s}}([0,1]^2 \setminus K) = 0$ and c_i has a version that is continuous on $[0,1]^2 \setminus K$.

Assumption 2.1 allows unbounded densities (as for Gaussian copulas near the corners), densities with jumps along finitely many curves (for instance when f is piecewise smooth, or when Y has finitely many atoms, each of which produces a horizontal strip), and singular parts concentrated on finitely many curves (for instance $Y = g(X_i)$ with g piecewise monotone).

Grid. For an integer $s \geq 1$ let $I_a = ((a-1)/s, a/s]$, $a = 1, \ldots, s$, with 0 assigned to I_1, and $Q_{ab} = I_a \times I_b$. The grid size is $s = \lceil N^{\alpha_g} \rceil$; the default is $\alpha_g = 1/2$. We write $\lambda_N = N/s^2$.

Definition 2.2 (statistics)**.** Let $a_j = \max\{1, \lceil sU_j \rceil\}$ be the row of observation j.

(a) Rank grid. Let R_j be the rank of Y_j in $Y_1, \ldots, Y_N$, ties being broken by order of appearance, and $b_j = \max\{1, \lceil (R_j - 1)s/N \rceil\}$. The occupancy matrix is $G_{ab} = \mathbf{1}\{\exists j : (a_j, b_j) = (a, b)\}$ and the *unfilled statistic* is $\hat{S}_N^{\circ} = s^{-2} \sum_{a,b} (1 - G_{ab})$.

(b) Moore fill. Let $\mathcal{N}_{ab}$ be the set of the (at most eight) cells $(a', b') \neq (a, b)$ of the grid with $|a' - a| \leq 1$, $|b' - b| \leq 1$. For a threshold $\theta_f \in (0, 1]$ set $\tilde{G}_{ab} = 1$ if $G_{ab} = 1$ or $|\mathcal{N}_{ab}|^{-1} \sum_{\mathcal{N}_{ab}} G_{a'b'} \geq \theta_f$, and $\tilde{G}_{ab} = 0$ otherwise. The fill uses G only (it does not cascade). The *adjusted ersatz* is $\hat{S}_N^{\mathrm{adj}} = s^{-2} \sum_{a,b} (1 - \tilde{G}_{ab})$; the default is $\theta_f = 1/2$.

(c) S-ersatz (Puy et al., 2024). With $s = \lceil \sqrt{N} \rceil$, the column is computed from the min-max scaled output, $b_j^{\mathrm{u}} = \max\{1, \lceil s(Y_j - \min_l Y_l)/(\max_l Y_l - \min_l Y_l) \rceil\}$, the occupancy matrix is $G_{ab}^{\mathrm{u}} = \mathbf{1}\{\exists j : (a_j, b_j^{\mathrm{u}}) = (a, b)\}$, and $\hat{S}_N^{\mathrm{ers}} = s^{-2} \sum_{a,b} (1 - G_{ab}^{\mathrm{u}})$, without fill. The scaling requires $\max_l Y_l > \min_l Y_l$, which almost surely holds for all large N unless Y is almost surely constant. When all outputs coincide, the implementation assigns every point to the first column. Puy et al. (2024) report the fraction of occupied cells, $1 - \hat{S}_N^{\mathrm{ers}}$; we use the fraction of empty cells throughout, so that for all three statistics larger values indicate a more influential input.

For an interior cell and $\theta_f = 1/2$ the fill rule reads "at least four of the eight neighbours are occupied"; on an edge (five neighbours) it reads "at least three", and in a corner (three neighbours) "at least two". The block of ranks assigned to column b is $B_b = \{r : \max\{1, \lceil (r-1)s/N \rceil\} = b\}$. Counting the integers $r - 1$ in $((b-1)N/s, bN/s]$ gives $|B_1| = \lfloor N/s \rfloor + 1$, $|B_s| = \lceil N/s \rceil - 1$ and $|B_b| \in \{\lfloor N/s \rfloor, \lceil N/s \rceil\}$ otherwise. In particular $\big||B_b| - N/s\big| \leq 1$ for every b, and every block is non-empty if $N > s$.

3 Exact finite-sample properties

Lemma 3.1 (ties)**.** *Under SRS, the joint law of the point set $\{(a_j, b_j)\}_{j \leq N}$ is the same whether ties among the Y_j are broken by order of appearance or uniformly at random. With random tie-breaking, the ranks of the Y_j coincide almost surely with the ranks of $V_j = F(Y_j-) + \zeta_j[F(Y_j) - F(Y_j-)]$, where $\zeta_1, \ldots, \zeta_N$ are independent $U(0,1)$ variables independent of the sample.*

Proof. If $Y_j < Y_l$ then $V_j \leq F(Y_j) \leq F(Y_l-) \leq V_l$, and equality of the extremes has probability zero. If $Y_j = Y_l$ then $V_j < V_l$ if and only if $\zeta_j < \zeta_l$. Hence ranking the V_j is ranking the Y_j and breaking ties with independent uniform marks, that is, at random. For the first claim, let π be a uniformly distributed permutation of $\{1, \ldots, N\}$, independent of the sample. The permuted sample $(U_{\pi(j)}, Y_{\pi(j)})_j$ has the same law as the original one, because the observations are independent and identically distributed. Breaking ties by order of appearance in the permuted sample breaks the ties of the original sample in the uniformly random order induced by π, and the point set $\{(a_j, b_j)\}$ does not otherwise depend on the labels. □

Consequently, for statements about the law of the statistics at a fixed N, we may and do assume under SRS that F is continuous, after replacing Y by V, and that the ranks are those of $V_1, \dots, V_N$, which are almost surely distinct. Almost sure statements along nested samples refer to the statistics computed from the ranks of the V_j, that is, with ties broken at random.

Proposition 3.2 (elementary properties)**.** *For every sample with $N > s$: (i) $0 \le \hat{S}_N^{\mathrm{adj}} \le \hat{S}_N^{\circ} \le 1 - 1/s$; (ii) $\hat{S}_N^{\circ} \ge 1 - N/s^2$; (iii) $\hat{S}_N^{\circ}$ and $\hat{S}_N^{\mathrm{adj}}$ are invariant under strictly increasing transformations of the output.*

Proof. (i) Column b contains the $|B_b| \ge 1$ observations whose ranks lie in B_b, hence at least one occupied cell, so at least s cells are occupied. The fill only turns empty cells into occupied ones, so $\tilde{G} \ge G$ cell by cell. (ii) At most N cells are occupied. (iii) Ranks are invariant. □

For $N = s$ the last block is empty and the upper bound in (i) can fail. Part (i) also shows that the Moore fill cannot increase the statistic. What the fill changes is the limit functional and, in particular, its value under independence (Sections 4 and 5).

Proposition 3.3 (histogram identity)**.** *Let $n_{ab} = \#\{j : (a_j, b_j) = (a, b)\}$ and let $\hat{P}_N(Q_{ab}) = n_{ab}/N$ be the empirical distribution of the rank-transformed sample on the grid. If $N \le s^2$ (in particular if $s = \lceil \sqrt{N} \rceil$), then, for any sampling design,*

$$\hat{S}_N^{\circ} = \frac{1}{2} \sum_{a,b} \left| \hat{P}_N(Q_{ab}) - \frac{1}{s^2} \right| = \max_A \left| \hat{P}_N(A) - \Pi(A) \right|, \tag{1}$$

where the maximum is over unions A of grid cells.

Proof. Since $\sum_{a,b}(n_{ab}/N - s^{-2}) = 0$, half the sum of absolute values equals the sum of the negative parts, $\sum_{a,b}(s^{-2} - n_{ab}/N)_+$. If $n_{ab} \ge 1$ then $n_{ab}/N \ge 1/N \ge s^{-2}$ and the term vanishes; if $n_{ab} = 0$ it equals s^{-2}. The second equality is the usual expression of the total variation distance between two probability vectors. □

The right-hand side of (1) is the plug-in histogram estimator of $\delta_i = \mathrm{TV}(C_i, \Pi)$ on the $s \times s$ grid, computed with the known uniform margins of the copula. It is the estimator of the measure of Borgonovo (2007) in its copula form (Proposition 5.2; Wei et al., 2014), which Plischke and Borgonovo (2019, Equation 30) place in niveau 3 of their classification; compare the given-data estimators of Plischke et al. (2013) and the Bernstein-copula estimator of Plischke and Borgonovo (2019, Section 4). Using the empirical margins $n_{a\cdot}n_{\cdot b}/N^2$ instead of s^{-2}, where $n_{a\cdot}$ and $n_{\cdot b}$ are the row and column totals, changes the right-hand side by at most $\frac{1}{2}\sum_a |n_{a\cdot}/N - 1/s| + \frac{1}{2}\sum_b |n_{\cdot b}/N - 1/s|$. The column term is at most $s/(2N) = O(N^{-1/2})$, because $n_{\cdot b} = |B_b|$ and $\big||B_b| - N/s\big| \le 1$. The row term is at most s/N for a Latin hypercube, whose rows contain $N/s + O(1)$ points, and is of order $(s/N)^{1/2} = O(N^{-1/4})$ under SRS.

Proposition 3.4 (finite-sample zero condition, fixed grid)**.** *Let $N \ge 2$ and let the sample be drawn by SRS. Let the columns be computed from the true V_j, $b_j^{\mathrm{or}} = \max\{1, \lceil sV_j \rceil\}$, and let $\hat{S}_N^{\mathrm{or}}$ be the resulting unfilled statistic. Then $\mathbb{E}\hat{S}_N^{\mathrm{or}} = s^{-2}\sum_{a,b}(1 - C_i(Q_{ab}))^N \ge (1 - s^{-2})^N$, with equality if and only if $C_i(Q_{ab}) = s^{-2}$ for all cells.*

Proof. A cell is empty with probability $(1 - C_i(Q_{ab}))^N$. The function $w \mapsto (1 - w)^N$ is strictly convex on $[0, 1]$ for $N \ge 2$, and $\sum_{a,b} C_i(Q_{ab}) = 1$; apply Jensen's inequality. □

The equality case identifies only the masses of the grid cells: every copula that gives mass s^{-2} to each cell, not only Π, attains the bound. Proposition 3.4 holds for every N but only for a grid whose cells are fixed in advance, whereas the columns of the rank grid are random. Section 4 gives the asymptotic statement that covers both.

4 Limits under simple random sampling

For $\lambda > 0$ and $\theta_f \in (0, 1]$ define

$$\Phi_\lambda(C) = \int_0^1 \int_0^1 e^{-\lambda c(u,v)}\, \mathrm{d}u\, \mathrm{d}v, \qquad \Psi_\lambda(C) = \int_0^1 \int_0^1 h_\lambda\big(c(u,v)\big)\, \mathrm{d}u\, \mathrm{d}v, \tag{2}$$

where c is the density of the absolutely continuous part of C and

$$h_\lambda(t) = \mathbb{P}\big[\nu_0 = 0\big]\, \mathbb{P}\Big[\textstyle\sum_{n=1}^{8} \mathbf{1}\{\nu_n > 0\} < 8\theta_f\Big] = e^{-\lambda t}\, \mathbb{P}\big[\,\mathrm{Bin}(8, 1 - e^{-\lambda t}) < 8\theta_f\big], \tag{3}$$

with $\nu_0, \ldots, \nu_8$ independent Poisson variables of mean λt. In words, $h_\lambda(t)$ is the probability that a cell is empty after the fill when its own count and the counts of its eight neighbours are independent Poisson variables of mean λt.

Theorem 4.1 (limits)**.** *Let Assumption 2.1 hold, let the sample be drawn by SRS, let the ranks be those of $V_1, \ldots, V_N$, and let $s = s_N \to \infty$ with $N/s_N^2 \to \lambda \in (0, \infty)$. Then, almost surely,*

$$\hat{S}_N^{\circ} \to \Phi_\lambda(C_i) \qquad \text{and} \qquad \hat{S}_N^{\mathrm{adj}} \to \Psi_\lambda(C_i).$$

Moreover, with $\mathcal{V}$ the vector of order statistics of $V_1, \ldots, V_N$ and every $t > 0$,

$$\mathbb{P}\big(|\hat{S}_N^{\circ} - \mathbb{E}[\hat{S}_N^{\circ} \mid \mathcal{V}]| \geq t\big) \leq 2e^{-2t^2 s^4/N}, \qquad \mathbb{P}\big(|\hat{S}_N^{\mathrm{adj}} - \mathbb{E}[\hat{S}_N^{\mathrm{adj}} \mid \mathcal{V}]| \geq t\big) \leq 2e^{-2t^2 s^4/(81N)}. \tag{4}$$

The same limits hold for the fixed grid of Proposition 3.4. If F has atoms and ties are broken by order of appearance, the statistics have at each N the same law as those computed from the ranks of the V_j (Lemma 3.1), so the convergence holds in probability. Almost sure convergence along nested samples is not established for this rule.

With $s = \lceil \sqrt{N} \rceil$ one has $N/s^2 \to 1$, so the default limits are Φ_1 and Ψ_1; since $s^4/N \sim N$, the bounds (4) decay exponentially in N. The proof uses three lemmas.

Lemma 4.2 (concomitants; Yang, 1977; David and Nagaraja, 2003, Section 6.8)**.** *Let (U_j, V_j), $j \leq N$, be independent copies of (U, V) with V continuous, let $V_{(1)} < \cdots < V_{(N)}$ be the order statistics, $U_{[r]}$ the U-value paired with $V_{(r)}$, and $v \mapsto P_v$ a regular conditional distribution of U given $V = v$. Conditionally on $\mathcal{V} = (V_{(1)}, \ldots, V_{(N)})$, the $U_{[r]}$ are independent and $U_{[r]}$ has law $P_{V_{(r)}}$.*

Proof. Conditionally on $(V_1, \ldots, V_N)$, the U_j are independent with $U_j \sim P_{V_j}$. The vector $(U_{[1]}, \ldots, U_{[N]})$ is obtained from $(U_1, \ldots, U_N)$ by the permutation that sorts the V_j, which is a function of $(V_1, \ldots, V_N)$. Hence, conditionally on $(V_1, \ldots, V_N)$, the $U_{[r]}$ are independent with $U_{[r]} \sim P_{V_{(r)}}$. This conditional law depends on $(V_1, \ldots, V_N)$ only through $\mathcal{V}$, which gives the claim. □

Lemma 4.3 (uniform closeness of order statistics)**.** *Let $d_N = \sqrt{\log N / N}$. Almost surely, for all N large enough, $\max_{r \leq N} |V_{(r)} - r/N| \leq d_N$.*

Proof. With F_N the empirical distribution function of the V_j, $|V_{(r)} - r/N| = |V_{(r)} - F_N(V_{(r)})| \leq \sup_v |F_N(v) - v|$. By the Dvoretzky–Kiefer–Wolfowitz inequality with Massart's constant (Massart, 1990), $\mathbb{P}(\sup_v |F_N(v) - v| > d_N) \leq 2e^{-2Nd_N^2} = 2N^{-2}$, which is summable; apply the Borel–Cantelli lemma. □

Lemma 4.4 (local Poisson limit)**.** *There is an event Ω_0 of probability one, which does not depend on (u, v), on which the following holds for every $(u, v) \in (0, 1)^2 \setminus K$. With $a = \lceil us \rceil$, $b = \lceil vs \rceil$ and $\mathfrak{B}$ the 3×3 block of cells $(a + \alpha, b + \beta)$, $\alpha, \beta \in \{-1, 0, 1\}$, the conditional law given $\mathcal{V}$ of the vector of counts $(n_{a'b'})_{(a',b') \in \mathfrak{B}}$ converges to the law of nine independent Poisson variables with mean $\lambda c_i(u, v)$.*

Proof. Disintegration. Let $O = [0,1]^2 \setminus K$, which is open, and let $I \times J$ be a product of intervals with rational endpoints (open, closed or half-open) whose closure lies in O. Because $C_i^{\mathrm{s}}(O) = 0$, every rational subinterval $J' \subset J$ satisfies $\int_{J'} P_v(I)\,\mathrm{d}v = C_i(I \times J') = \int_{J'}\int_I c_i(u,v)\,\mathrm{d}u\,\mathrm{d}v$, so $P_v(I) = \int_I c_i(u,v)\,\mathrm{d}u$ for every $v \in J$ outside a Lebesgue-null set. Let $\mathcal{N}_0$ be the union of these null sets over the countably many such rectangles. If $v \notin \mathcal{N}_0$ and I is an interval with rational endpoints and $\bar{I} \times \{v\} \subset O$, compactness gives an open interval $J \ni v$ with rational endpoints and $\bar{I} \times \bar{J} \subset O$, hence $P_v(I) = \int_I c_i(u,v)\,\mathrm{d}u$. Let Ω_0 be the intersection of the event of Lemma 4.3 with the event that no V_j falls in $\mathcal{N}_0$; since the V_j are continuous, Ω_0 has probability one.

Localisation. For $r \in B_{b'}$ we have $(r-1)s/N \in (b'-1, b']$ if $b' \geq 2$ and $(r-1)s/N \leq 1$ if $b' = 1$, so on Ω_0 and for N large, $|V_{(r)} - v| \leq 2/s + 1/N + d_N$ whenever $|b' - b| \leq 1$. Let $\rho_N = 3/s + d_N \to 0$ and $D_N = [u - \rho_N, u + \rho_N] \times [v - \rho_N, v + \rho_N]$. Because K is closed and $(u,v) \notin K$, $D_N \cap K = \emptyset$ for N large, and by continuity $\eta_N = \sup_{D_N} |c_i - c_i(u,v)| \to 0$. For such N, every row interval $I_{a'}$ of the block (a half-open interval with rational endpoints) and every $V_{(r)}$ with $r \in B_{b-1} \cup B_b \cup B_{b+1}$ satisfy $\bar{I}_{a'} \times \{V_{(r)}\} \subset D_N \subset O$, hence

$$\pi_{r,a'} := P_{V_{(r)}}(I_{a'}) = \int_{I_{a'}} c_i(u', V_{(r)})\,\mathrm{d}u', \qquad |s\,\pi_{r,a'} - c_i(u,v)| \leq \eta_N.$$

Generating function. By Lemma 4.2, each concomitant with rank in $B_{b'}$ falls independently in row a' with probability $\pi_{r,a'}$, and in at most one row. For $z = (z_{a'b'}) \in [0,1]^9$,

$$\mathbb{E}\Big[\prod_{\mathfrak{B}} z_{a'b'}^{n_{a'b'}} \,\Big|\, \mathcal{V}\Big] = \prod_{b'=b-1}^{b+1} \prod_{r \in B_{b'}} \Big(1 + \sum_{a'=a-1}^{a+1} \pi_{r,a'}(z_{a'b'} - 1)\Big).$$

Write x_r for the inner sum. Then $\max_r |x_r| \leq 3(c_i(u,v) + \eta_N)/s \to 0$, and $\sum_{r \in B_{b'}} \pi_{r,a'} = |B_{b'}|(c_i(u,v) + O(\eta_N))/s \to \lambda c_i(u,v)$, because $|B_{b'}| = N/s + O(1)$ and $N/s^2 \to \lambda$. Since $\log(1+x) = x + O(x^2)$ and $\sum_r x_r^2 \leq \max_r |x_r| \sum_r |x_r| \to 0$, the generating function converges to $\prod_{\mathfrak{B}} \exp\{\lambda c_i(u,v)(z_{a'b'} - 1)\}$, the generating function of nine independent Poisson variables with mean $\lambda c_i(u,v)$. Convergence of generating functions on $[0,1]^9$ implies convergence in distribution of $\mathbb{N}^9$-valued vectors; see Kolchin et al. (1978) for Poisson limits in allocation problems of this kind. □

Proof of Theorem 4.1. Write $\hat{S}_N^{\circ} = \int_0^1\int_0^1 \mathbf{1}\{n_{a_N(u)b_N(v)} = 0\}\,\mathrm{d}u\,\mathrm{d}v$ with $a_N(u) = \lceil us \rceil$, $b_N(v) = \lceil vs \rceil$, and similarly $\hat{S}_N^{\mathrm{adj}}$ with the indicator that the cell is empty after the fill. Fix $(u,v) \in (0,1)^2 \setminus K$. For N large the cell $(a_N(u), b_N(v))$ is interior, and both indicators are functions of the nine counts of Lemma 4.4: the first is $\mathbf{1}\{n_{ab} = 0\}$, the second is $\mathbf{1}\{n_{ab} = 0\}\mathbf{1}\{\sum_{\mathcal{N}_{ab}} \mathbf{1}\{n_{a'b'} > 0\} < 8\theta_f\}$. As the counts take values in a discrete set, Lemma 4.4 gives, on Ω_0,

$$\mathbb{P}(n_{ab} = 0 \mid \mathcal{V}) \to e^{-\lambda c_i(u,v)}, \qquad \mathbb{P}(\text{cell empty after fill} \mid \mathcal{V}) \to h_\lambda(c_i(u,v)).$$

Since $\mathrm{Leb}(K) = 0$ this holds for almost every (u,v), and the integrands are bounded by one; because Ω_0 does not depend on (u,v), Fubini's theorem and dominated convergence give $\mathbb{E}[\hat{S}_N^{\circ} \mid \mathcal{V}] \to \Phi_\lambda(C_i)$ and $\mathbb{E}[\hat{S}_N^{\mathrm{adj}} \mid \mathcal{V}] \to \Psi_\lambda(C_i)$ on Ω_0.

For the fluctuations, conditionally on $\mathcal{V}$ the statistics are functions of the independent concomitants $U_{[1]}, \ldots, U_{[N]}$ (Lemma 4.2), and the column of each concomitant is fixed by its rank. Changing one concomitant moves one point within its column, which can be done by removing the point and then adding it in its new cell. Removing it can only empty a cell of G, and adding it can only fill one, so the number of empty cells of G changes by at most one in each step, in opposite directions, and $\hat{S}_N^{\circ}$ changes by at most s^{-2}. The fill map $G \mapsto \tilde{G}$ is monotone ($G \leq G'$ cell by cell implies $\tilde{G} \leq \tilde{G}'$), and a change of G in one cell can alter $\tilde{G}$ only in that cell and its at most eight neighbours. Hence the removal can only empty cells of $\tilde{G}$, at most nine, and the addition can only fill cells of $\tilde{G}$, at most nine, so $\hat{S}_N^{\mathrm{adj}}$ changes by at most $9s^{-2}$.

McDiarmid's bounded-difference inequality (McDiarmid, 1989), applied conditionally on $\mathcal{V}$, gives (4). As $s^4/N \to \infty$ proportionally to N, the right-hand sides are summable for every $t > 0$, and the Borel–Cantelli lemma yields $\hat{S}_N^{\circ} - \mathbb{E}[\hat{S}_N^{\circ} \mid \mathcal{V}] \to 0$ and $\hat{S}_N^{\mathrm{adj}} - \mathbb{E}[\hat{S}_N^{\mathrm{adj}} \mid \mathcal{V}] \to 0$ almost surely.

For the fixed grid the columns are the intervals I_b, and the concomitants in column b are those with $V_{(r)} \in I_b$. Their number is binomial with mean N/s; by Bernstein's inequality and a union bound over the s columns it equals $(N/s)(1 + o(1))$ uniformly in b, almost surely, and the same argument applies. □

Corollary 4.5 (consistency of the ranking and of screening)**.** *Let the assumptions of Theorem 4.1 hold for every input.*

(i) If $\Psi_\lambda(C_1), \dots, \Psi_\lambda(C_k)$ are pairwise distinct, then almost surely, for all N large enough, the ranking of the inputs by $\hat{S}_N^{\mathrm{adj}}$ coincides with their ranking by $\Psi_\lambda(C_i)$. The same holds for $\hat{S}_N^{\circ}$ and Φ_λ.

(ii) Let $\theta_f = 1/2$ and $\lambda \geq 3/4$, and let $\tau_N \to 0$ with $\tau_N (N/\log N)^{1/2} \to \infty$ (for instance $\tau_N = N^{-1/4}$). The test that declares X_i influential if $\hat{S}_N^{\mathrm{adj}} > \kappa_{\lambda_N} + \tau_N$ is consistent against every fixed alternative: almost surely, for all N large enough, it declares influential every input with $C_i \neq \Pi$ and non-influential every input independent of Y.

Proof. (i) follows from the almost sure convergence of finitely many statistics to distinct limits.

(ii) Alternatives. If $C_i \neq \Pi$, then $\Psi_\lambda(C_i) > \kappa_\lambda = h_\lambda(1)$ by Proposition 5.4(ii), while $\kappa_{\lambda_N} + \tau_N \to \kappa_\lambda$ by continuity of $\lambda \mapsto h_\lambda(1)$. Since $\hat{S}_N^{\mathrm{adj}} \to \Psi_\lambda(C_i)$ almost surely, the input is eventually declared influential.

Independence. If X_i and Y are independent, the concomitants are independent and uniform on $(0, 1)$ whatever the value of $\mathcal{V}$, and the block sizes are deterministic, so $\mu_N = \mathbb{E}[\hat{S}_N^{\mathrm{adj}} \mid \mathcal{V}]$ is a number. We show that $|\mu_N - \kappa_{\lambda_N}| \leq (13 + 27\lambda_N)/s + 27/s^2$, a bound that is informative only for large s and is used here only asymptotically. The $4s - 4$ boundary cells contribute at most $4/s$. For an interior cell, the counts of its 3×3 block are, column by column, the counts in three of the s rows of $|B_{b'}|$ independent uniform points, and the three columns are independent. Each point falls in these three rows with probability $3/s$. Given the number of points that fall in the three rows, both the multinomial counts and three independent Poisson variables with mean $|B_{b'}|/s$ split these points equally likely among the rows, so the total variation distance between the two laws equals that between $\mathrm{Bin}(|B_{b'}|, 3/s)$ and the Poisson law with mean $3|B_{b'}|/s$. By Le Cam's inequality (Barbour et al., 1992, Chapter 1) it is at most $|B_{b'}|(3/s)^2 \leq 9\lambda_N/s + 9/s^2$. Since $\big||B_{b'}|/s - \lambda_N\big| \leq 1/s$, each of these Poisson laws is within total variation $1/s$ of the Poisson law with mean λ_N. The probability that the cell is empty after the fill is therefore within $27\lambda_N/s + 27/s^2 + 9/s$ of its value $\kappa_{\lambda_N} = h_{\lambda_N}(1)$ for nine independent Poisson(λ_N) counts, which proves the bound. Since $s^2 \sim N/\lambda$ and $\tau_N\sqrt{N} \to \infty$, $|\mu_N - \kappa_{\lambda_N}| \leq \tau_N/2$ for N large. Then, by the one-sided form of (4),

$$\mathbb{P}\big(\hat{S}_N^{\mathrm{adj}} > \kappa_{\lambda_N} + \tau_N\big) \leq \mathbb{P}\big(\hat{S}_N^{\mathrm{adj}} - \mu_N \geq \tau_N/2\big) \leq \exp\{-\tau_N^2 s^4/(162N)\},$$

and since $s^4/N \sim N/\lambda^2$ and $\tau_N^2 N/\log N \to \infty$, the right-hand side is summable. The Borel–Cantelli lemma concludes. □

This is the precise sense in which the adjusted ersatz is consistent: it is a strongly consistent estimator of the functional $\Psi_\lambda(C_i)$ when the output distribution is continuous, a weakly consistent one (convergence in probability) when the output has atoms and ties are broken by order of appearance, and it yields a consistent test for independence. In the latter case (i) and (ii) hold with probability tending to one, instead of almost surely for all N large enough, because the statistics have the same law at each N as with random tie-breaking (Lemma 3.1). The adjusted ersatz is not a consistent estimator of δ_i or of T_i, which are different functionals.

5 The limit functionals

Recall that the f-divergence of a copula C from Π, for convex $f : [0,\infty) \to \mathbb{R}$ with $f(1) = 0$, is $D_f(C\|\Pi) = \int f(c)\,\mathrm{d}\Pi + m\, f'(\infty)$, where $f'(\infty) = \lim_{t\to\infty} f(t)/t$ (Csiszár, 1967; Liese and Vajda, 2006). For the copula C_i of Section 2, $D_f(C_i\|\Pi)$ equals the f-information $I_f(X_i;Y) = D_f(P_{X_iY}\|P_{X_i}\otimes P_Y)$. Indeed (X_i,V) is obtained from (X_i,Y) by a Markov kernel acting on the second coordinate (the distributional transform), and $(X_i,Y) = (X_i, F^{-1}(V))$ almost surely; the same kernels map $P_{X_i}\otimes P_Y$ to $P_{X_i}\otimes P_V = \Pi$ and back, so the data-processing inequality (Liese and Vajda, 2006) applies in both directions. Moreover $I_f(X_i;Y) = \mathbb{E}_{X_i}[D_f(P_{Y|X_i}\|P_Y)]$, which is the f-sensitivity index of Rahman (2016); see Borgonovo et al. (2016) and Da Veiga (2015) for the common rationale of such dependence-based measures. Borgonovo's measure corresponds to $f(t) = |t-1|/2$ (Borgonovo, 2007; Borgonovo et al., 2011), and mutual information to $f(t) = t\log t$. For absolutely continuous C, $D_f(C\|\Pi)$ is the measure δ_h of Plischke and Borgonovo (2019, Equation 31) with $h(t) = f(t)/t$ for $t > 0$, which satisfies their conditions $h(1) = 0$ and $t \mapsto h(1/t)$ convex, so these indices belong to their niveau 3.

Proposition 5.1 (Φ_λ is an f-divergence)**.** *Let $f_\lambda(t) = e^{-\lambda t} - e^{-\lambda}$, which is strictly convex with $f_\lambda(1) = 0$ and $f'_\lambda(\infty) = 0$. For every copula C, $\Phi_\lambda(C) - e^{-\lambda} = D_{f_\lambda}(C\|\Pi)$. Consequently:*

(i) $e^{-\lambda} \le \Phi_\lambda(C) \le 1$; $\Phi_\lambda(C) = e^{-\lambda}$ if and only if $C = \Pi$, and $\Phi_\lambda(C) = 1$ if and only if C is singular;

(ii) $\Phi_\lambda(C_i) - e^{-\lambda} = I_{f_\lambda}(X_i;Y)$, the f_λ-sensitivity index of X_i; in particular it is invariant under strictly increasing transformations of X_i and of Y;

(iii) (data processing) if X_i' is obtained from X_i through a Markov kernel, independently of Y given X_i, then $I_{f_\lambda}(X_i';Y) \le I_{f_\lambda}(X_i;Y)$, and similarly for transformations of the output.

Proof. The identity is the definition of D_{f_λ} with $f'_\lambda(\infty) = 0$. (i) By Jensen's inequality, $\int e^{-\lambda c} \ge \exp(-\lambda\int c) = e^{-\lambda(1-m)} \ge e^{-\lambda}$. Equality in the second step requires $m = 0$, and in the first, by strict convexity, c constant; hence $c \equiv 1$ and $C = \Pi$. The upper bound holds because $e^{-\lambda c} \le 1$, with equality only where $c = 0$. (ii) is the identity recalled before the proposition. (iii) is the data-processing inequality for f-divergences (Csiszár, 1967; Liese and Vajda, 2006), applied to the joint law and to the product of the margins, which are transformed by the same kernel. □

Proposition 5.2 (Φ_λ and δ)**.** *Let $\delta(C) = \mathrm{TV}(C,\Pi) = \sup_A |C(A) - \Pi(A)|$. Then $\delta(C) = \int (1-c)_+\,\mathrm{d}\Pi$, and $\delta(C_i) = \delta_i$, the measure of Borgonovo (2007), which equals $\frac{1}{2}\mathbb{E}_{X_i}\int |f_Y(y) - f_{Y|X_i}(y)|\,\mathrm{d}y$ when densities exist. For every $\lambda > 0$ and every copula C,*

$$\Phi_\lambda(C) - e^{-\lambda} \le (1 - e^{-\lambda})\,\delta(C),$$

and the constant $1 - e^{-\lambda}$ is sharp. If C is absolutely continuous with $c \le \bar{c}$, then $\Phi_\lambda(C) - e^{-\lambda} \ge 2\lambda^2 e^{-\lambda\bar{c}}\,\delta(C)^2$. For $\lambda = 1$: $2e^{-\bar{c}}\delta^2 \le \Phi_1 - e^{-1} \le (1 - e^{-1})\,\delta$.

Proof. Let $\mathcal{S}$ be a Lebesgue-null set with $C^{\mathrm{s}}([0,1]^2\setminus\mathcal{S}) = 0$. For every Borel set A, $\Pi(A) - C(A) = \int_A (1-c)\,\mathrm{d}\Pi - C^{\mathrm{s}}(A) \le \int (1-c)_+\,\mathrm{d}\Pi$, with equality for $A = \{c < 1\}\setminus\mathcal{S}$. Since both measures have mass one, $\sup_A(C(A) - \Pi(A)) = \sup_A(\Pi(A) - C(A))$, which proves the first claim. The total variation distance is the f-divergence with $f(t) = |t-1|/2$, so $\delta(C_i) = \mathrm{TV}(P_{X_iY}, P_{X_i}\otimes P_Y)$ by the identity recalled at the beginning of this section, and this is Borgonovo's definition (Borgonovo, 2007; Plischke and Borgonovo, 2019).

Upper bound. f_λ is convex and decreasing with $f_\lambda(1) = 0$, so $f_\lambda(t) \le (1-t)f_\lambda(0) = (1-t)(1-e^{-\lambda})$ for $t \in [0,1]$ (the chord lies above the graph) and $f_\lambda(t) \le 0$ for $t \ge 1$. As $f'_\lambda(\infty) = 0$, $D_{f_\lambda}(C\|\Pi) = \int f_\lambda(c)\,\mathrm{d}\Pi \le (1-e^{-\lambda})\int (1-c)_+\,\mathrm{d}\Pi$. For sharpness, let $a \in (0,1)$ and let C_a have density 0 on the square $[0,a]^2$, density 1 elsewhere, and a singular part of

mass a^2 spread uniformly on the diagonal of $[0,a]^2$. Its margins are uniform, $\delta(C_a) = a^2$ and $\Phi_\lambda(C_a) - e^{-\lambda} = a^2(1-e^{-\lambda})$.

Lower bound. Let $\bar f(t) = f_\lambda(t) + \lambda e^{-\lambda}(t-1)$, which is convex with $\bar f(1) = \bar f'(1) = 0$. If C is absolutely continuous, $\int (c-1)\,\mathrm{d}\Pi = 0$ and $D_{f_\lambda}(C\|\Pi) = \int \bar f(c)\,\mathrm{d}\Pi$. On $[0,\bar c]$, $\bar f''(t) = \lambda^2 e^{-\lambda t} \ge \lambda^2 e^{-\lambda \bar c}$, so $\bar f(t) \ge \frac{1}{2}\lambda^2 e^{-\lambda\bar c}(t-1)^2$, and by the Cauchy–Schwarz inequality $D_{f_\lambda} \ge \frac{1}{2}\lambda^2 e^{-\lambda\bar c}(\int |c-1|\,\mathrm{d}\Pi)^2 = 2\lambda^2 e^{-\lambda\bar c}\delta^2$. □

Remark 5.3 (the histogram estimator of δ at critical resolution). Propositions 3.3, 5.2 and Theorem 4.1 together answer the question whether the rank grid estimates δ. It is the histogram estimator of δ, but when $N/s^2 \to \lambda \in (0,1]$, the regime of the ersatz, it converges to $\Phi_\lambda(C_i)$. Since $e^{-\lambda t} \ge e^{-t} \ge (1-t)_+$ with equality only at $t=0$, $\Phi_\lambda(C_i) \ge \delta_i$ with equality only for singular C_i; under independence the limit is $e^{-\lambda}$ instead of 0. Consistency for δ requires $N/s^2 \to \infty$ (Devroye and Györfi, 1985), a regime in which the identity (1) no longer holds and the empty-cell fraction has a different limit (Theorem 6.1).

Proposition 5.4 (properties of Ψ_λ; default $\theta_f = 1/2$)**.** *Let $\varphi(p) = p^6(56 - 140p + 120p^2 - 35p^3)$. Then $h_\lambda(t) = \varphi(e^{-\lambda t})$, and:*

(i) φ is strictly increasing on $[0,1]$ with $\varphi(0) = 0$ and $\varphi(1) = 1$; hence h_λ is strictly decreasing, $h_\lambda(0) = 1$, and $\Psi_\lambda(C) \le 1$ with equality if and only if C is singular. The value under independence is $\kappa_\lambda := \Psi_\lambda(\Pi) = \varphi(e^{-\lambda})$; $\kappa_1 = 0.04708$.

(ii) (zero condition; constants computed in 40-digit arithmetic) For every $\lambda \ge 3/4$, $h_\lambda(t) \ge h_\lambda(1) + h'_\lambda(1)(t-1)$ for all $t \ge 0$, with equality only at $t = 1$. Consequently $\Psi_\lambda(C) \ge \kappa_\lambda$ for every copula C, with equality if and only if $C = \Pi$.

(iii) (fine grids) Conversely, if the inequality of (ii) fails at some $t \ne 1$, there is a copula with $\Psi_\lambda(C) < \kappa_\lambda$. The inequality fails for every $\lambda \in (-\log p_1, \lambda_c) = (0.0422, 0.4815)$, where p_1 is defined in the proof of (ii) and $\lambda_c = 0.481480$ is the root of $L(e^{-\lambda}) = 1$, with $L(p) = \varphi(p) - p\log(p)\,\varphi'(p)$. On $[\lambda_c, 3/4)$ the inequality holds (computer-assisted), with equality only at $t = 1$ for $\lambda > \lambda_c$ and only at $t \in \{0,1\}$ for $\lambda = \lambda_c$, and the conclusion of (ii) holds there too. For $\lambda > 0.0422$ the bound $\Psi_\lambda \ge \kappa_\lambda$ therefore holds exactly when $\lambda \ge \lambda_c$.

(iv) (approximation of δ; upper bound computer-assisted) For every $\lambda > 0$ and every copula C, $\Psi_\lambda(C) \ge \delta(C) - \varepsilon_\lambda$, where $\varepsilon_\lambda := \sup_{0\le t\le 1}\big[(1-t) - h_\lambda(t)\big]$, and for $\lambda \in [1/2, 1]$, $\Psi_\lambda(C) \le \delta(C) + \kappa_\lambda$. Both bounds are sharp. Numerically $\varepsilon_1 = 0.14003$ (attained at $t = 0.586$), so that $\delta(C) - 0.1401 \le \Psi_1(C) \le \delta(C) + 0.0471$, and $\varepsilon_\lambda = 0$ for $\lambda \le 0.7925$; for the grid used at $N = 512$ ($s = 23$, $\lambda = 512/529$), $\kappa = 0.0548$ and $\varepsilon = 0.1208$.

Proof. For $\theta_f = 1/2$, $h_\lambda(t) = p\,\mathbb{P}[\mathrm{Bin}(8, 1-p) \le 3]$ with $p = e^{-\lambda t}$, and $\mathbb{P}[\mathrm{Bin}(8,1-p) \le 3] = \sum_{n=5}^{8}\binom{8}{n}p^n(1-p)^{8-n}$; expanding gives φ.

(i) $\varphi'(p) = p^5 r_3(p)$ with $r_3(p) = 336 - 980p + 960p^2 - 315p^3$. The derivative $r_3'(p) = -980 + 1920p - 945p^2$ has negative discriminant (-18000), so r_3 is strictly decreasing, and $r_3(1) = 1 > 0$; hence $\varphi' > 0$ on $(0,1]$. The other claims follow, using that $\Psi_\lambda(C) = 1$ forces $h_\lambda(c) = 1$, that is $c = 0$, almost everywhere.

(ii) Write $p = e^{-\lambda t}$. Then $h''_\lambda(t) = \lambda^2 p\, q(p)$ with $q(p) = \frac{\mathrm{d}}{\mathrm{d}p}[p\,\varphi'(p)] = p^5(2016 - 6860p + 7680p^2 - 2835p^3)$. The cubic factor has the real roots $p_0 = 0.719781\ldots$, $p_1 = 0.958720\ldots$ and $p_2 = 1.030494\ldots$ (exact real-root isolation) and is positive on $(0, p_0)$. Hence h_λ is strictly convex on $[t_0, \infty)$ with $t_0 = -\log(p_0)/\lambda$. For $\lambda \ge 3/4$, $e^{-\lambda} \le e^{-3/4} = 0.472 < p_0$, so $t = 1$ lies in this interval and the tangent line $\ell(t) = h_\lambda(1) + h'_\lambda(1)(t-1)$ lies strictly below h_λ on $[t_0,\infty)\setminus\{1\}$. On $[0, t_0)$, $h_\lambda(t) > h_\lambda(t_0) = \varphi(p_0) = 0.604780$, while ℓ is decreasing, so $\ell(t) \le \ell(0) = L(e^{-\lambda})$. Since $L'(p) = -\log(p)\,q(p) > 0$ on $(0,p_0)$, $L(e^{-\lambda}) \le L(e^{-3/4}) = 0.594543 < 0.604780$. This proves the tangent inequality. Integrating it, $\Psi_\lambda(C) = \int h_\lambda(c)\,\mathrm{d}\Pi \ge h_\lambda(1) + h'_\lambda(1)(\int c\,\mathrm{d}\Pi - 1) = \kappa_\lambda - h'_\lambda(1)m \ge \kappa_\lambda$, because $h'_\lambda(1) < 0$; equality forces $m = 0$ and $c = 1$ almost everywhere.

(iii) For every probability density ϱ on $[0,1]$, $c(u,v) = \varrho((u+v) \bmod 1)$ is the density of a copula, a circulant copula, because each of its sections integrates to one. Let $t_1 \neq 1$ with $h_\lambda(t_1) < \ell(t_1)$, and let ϱ take the value t_1 on a set of measure w and a value t_2 on the other side of 1 elsewhere, with $wt_1 + (1-w)t_2 = 1$. Because ℓ is affine with $\ell(1) = \kappa_\lambda$,

$$\Psi_\lambda - \kappa_\lambda = w\,[h_\lambda(t_1) - \ell(t_1)] + (1-w)\,[h_\lambda(t_2) - \ell(t_2)].$$

As $t_2 \to 1$, $w = (t_2-1)/(t_2-t_1)$ is of order $|t_2-1|$ while the second bracket is of order $(t_2-1)^2$, so $\Psi_\lambda < \kappa_\lambda$ once $|t_2-1|$ is small enough. A simpler example at $\lambda = 0.2268$ (the value of λ_N for $N = 1600$ and $\alpha_g = 0.6$) is the band copula with density 2 on $\{(u+v) \bmod 1 < 1/2\}$ and 0 elsewhere: $\delta = 0.5$ and $\Psi_\lambda = 0.714 < \kappa_\lambda = 0.750$. For $\lambda \in [-\log p_0, \lambda_c)$, $e^{-\lambda}$ lies in $(e^{-\lambda_c}, p_0]$, where L is increasing, so $\ell(0) = L(e^{-\lambda}) > L(e^{-\lambda_c}) = 1 = h_\lambda(0)$ and the inequality fails at $t = 0$. For $\lambda \in (-\log p_1, -\log p_0)$, $e^{-\lambda} \in (p_0, p_1)$, where the cubic factor of q is negative, so $h_\lambda''(1) < 0$ and h_λ lies strictly below its tangent on a punctured neighbourhood of $t = 1$.

For $\lambda \in [\lambda_c, 3/4)$ write $D(t) = h_\lambda(t) - \ell(t)$, where $\ell(t) = \kappa_\lambda + \lambda\psi(e^{-\lambda})(1-t)$ and $\psi(p) = p\,\varphi'(p)$, so that $h_\lambda'(t) = -\lambda\psi(e^{-\lambda t})$. Since $\psi' = q$, ψ is increasing on $(0, p_0)$ and on $(p_1, 1]$, with $\psi(1) = 1$. As in (ii), h_λ is strictly convex on $[t_0, \infty)$, where $t_0 = -\log(p_0)/\lambda \leq 0.683 < 1$, so $D > 0$ there except at $t = 1$, and it remains to show $D \geq 0$ on $[0, t_0)$. (a) For $\lambda \leq 0.55$ and $t \leq 0.07$, $e^{-\lambda t} \geq e^{-0.0385} > p_1$, so $\psi(e^{-\lambda t}) \leq 1$, while $\psi(e^{-\lambda}) \geq \psi(e^{-0.55}) = 1.0936$. Hence $D'(t) = \lambda[\psi(e^{-\lambda}) - \psi(e^{-\lambda t})] > 0$, and $D(t) \geq D(0) = 1 - L(e^{-\lambda}) \geq 0$, with equality only for $t = 0$ and $\lambda = \lambda_c$, because L is increasing on $(0, p_0)$. (b) On the rest of $[0, t_0) \times [\lambda_c, 3/4)$ the inequality was verified by monotone bracketing, as in (iv): for $t \in [t_a, t_b]$ and $\lambda \in [\lambda_1, \lambda_2]$, $h_\lambda(t) \geq \varphi(e^{-\lambda_2 t_b})$ and $\ell(t) \leq \varphi(e^{-\lambda_1}) + \lambda_2\psi(e^{-\lambda_1})(1-t_a)$, because φ is increasing and $e^{-\lambda_1} < p_0$, and the difference of the bounds is positive (at least 0.0076) on a partition into 161,897 cells of size $10^{-3} \times 10^{-3}$. Integrating as in (ii) gives $\Psi_\lambda(C) \geq \kappa_\lambda$, with equality only if $m = 0$ and $c \in \{0, 1\}$ almost everywhere (only $c = 1$ if $\lambda > \lambda_c$). Since each section of the density of an absolutely continuous copula integrates to one, $c \in \{0, 1\}$ forces $c = 1$ almost everywhere, so equality holds only for $C = \Pi$, also at $\lambda = \lambda_c$.

(iv) By Proposition 5.2, $\Psi_\lambda(C) - \delta(C) = \int [h_\lambda(c) - (1-c)_+]\,\mathrm{d}\Pi$, so it suffices to bound the integrand. The lower bound holds for every $\lambda > 0$ by the definition of ε_λ on $[0, 1]$ and because $h_\lambda \geq 0 = (1-t)_+$ on $(1, \infty)$. For the upper bound, $h_\lambda(t) \leq h_\lambda(1) = \kappa_\lambda$ for $t \geq 1$ by monotonicity. On $[0, 1]$ the inequality $h_\lambda(t) \leq 1 - t + \kappa_\lambda$ was verified by monotone bracketing: for $t \in [t_a, t_b]$ and $\lambda \in [\lambda_1, \lambda_2]$ one has $h_\lambda(t) \leq \varphi(e^{-\lambda_1 t_a})$ and $1 - t + \kappa_\lambda \geq 1 - t_b + \varphi(e^{-\lambda_2})$, and the difference of the bounds is negative (at most -0.006) on a partition of $[0, 0.98] \times [1/2, 1]$ into cells of size $5 \cdot 10^{-5} \times 10^{-3}$. On $[0.98, 1]$ the function $t \mapsto 1 - t + \kappa_\lambda - h_\lambda(t)$ vanishes at $t = 1$ and is decreasing, because $|h_\lambda'(t)| = \lambda\psi(e^{-\lambda t}) < 1$ with $\psi(p) = p\varphi'(p)$: ψ is increasing on $(0, p_0)$, since $\psi' = q$, and $e^{-\lambda t} \leq e^{-0.49} < p_0$ there, so bounding λ by the upper end and $e^{-\lambda t}$ by its value at the lower ends of the same cells gives $|h_\lambda'| \leq 0.6231$. The upper bound is attained at $C = \Pi$. The lower bound is approached by circulant copulas whose density equals the maximiser t^* of $(1-t) - h_\lambda(t)$ on a set of measure w and a large value t_2 elsewhere, with $wt^* + (1-w)t_2 = 1$: then $\Psi_\lambda - \delta = w[h_\lambda(t^*) - (1-t^*)] + (1-w)h_\lambda(t_2) \to -\varepsilon_\lambda$ as $t_2 \to \infty$. □

Remark 5.5 (what the fill does). Figure 5(a) of the main text displays the three integrands $(1-t)_+$, e^{-t} and $h_1(t)$. The rank grid without fill converges to a functional that is compressed into $[e^{-1}, 1]$ and exceeds δ by up to $e^{-1} \approx 0.37$ (Remark 5.3). The Moore fill is a local denoising step: an empty cell surrounded by occupied cells is the typical outcome of sampling variability in a region of positive density, and it is filled, while an empty cell in an empty region is kept. In the limit this replaces $e^{-\lambda t}$ by $h_\lambda(t)$, a smooth surrogate of the integrand $(1-t)_+$ of δ. For Gaussian copulas, numerically, $|\Psi_1 - \delta| \leq 0.0471$, the maximum being attained at independence, and $\delta - \Psi_1 \leq 0.023$, with the maximum near correlation 0.61 (240 values of the correlation in $[0, 0.999]$; Section 10).

Proposition 5.6 (the S-ersatz depends on the output distribution)**.** *Let the sample be drawn by SRS, with* $s = \lceil \sqrt{N} \rceil$.

(i) Let X_i and Y be independent and let Y have bounded support $[y_{\mathrm{lo}}, y_{\mathrm{hi}}]$ and a density f_Y that is continuous and positive on $(y_{\mathrm{lo}}, y_{\mathrm{hi}})$. Let $\tilde{f}(v) = (y_{\mathrm{hi}} - y_{\mathrm{lo}})\, f_Y(y_{\mathrm{lo}} + v(y_{\mathrm{hi}} - y_{\mathrm{lo}}))$ be the density of the min-max scaled output. Then $\hat{S}_N^{\mathrm{ers}} \to \int_0^1 e^{-\tilde{f}(v)}\,\mathrm{d}v$ almost surely, and the limit is at least e^{-1}, with equality if and only if Y is uniform on its support.

(ii) If the support of Y is unbounded, then $\hat{S}_N^{\mathrm{ers}} \to 1$ almost surely, whatever the dependence between X_i and Y.

Proof. The sample minimum and maximum converge almost surely to y_{lo} and y_{hi}. Conditionally on the indices and values of the two extreme observations, the other $N-2$ outputs are independent with the law of Y restricted to $(\min_l Y_l, \max_l Y_l)$, and independent of their U_j. (i) A cell (a, b) therefore receives each of these points independently with probability $s^{-1}\varpi_b$, where ϖ_b is the conditional probability of the b-th of s equal subintervals of $(\min_l Y_l, \max_l Y_l)$. For fixed $v \in (0, 1)$ and $b = \lceil vs \rceil$, continuity of f_Y gives $s\,\varpi_b \to \tilde{f}(v)$ almost surely, so the conditional probability that the cell is empty tends to $e^{-\tilde{f}(v)}$. Dominated convergence and the bounded-difference argument of Theorem 4.1 (the two extreme points change the count by at most $2s^{-2}$) complete the proof of the limit. Jensen's inequality gives $\int e^{-\tilde{f}} \geq e^{-\int \tilde{f}} = e^{-1}$, with equality if and only if $\tilde{f} \equiv 1$.

(ii) Suppose, say, that the support of Y is unbounded above. Then $\max_l Y_l \to \infty$ almost surely, so the range $\rho_N = \max_l Y_l - \min_l Y_l$ diverges. Fix $M > 0$. The columns of the grid are intervals of length ρ_N/s of the output scale, so at most $2Ms/\rho_N + 2 = o(s)$ columns meet $[-M, M]$, and they contain at most $o(s^2)$ cells. Every other occupied cell contains a point with $|Y_j| > M$, and by the strong law of large numbers there are at most $N\{\mathbb{P}(|Y| > M) + o(1)\}$ such points. Hence the fraction of occupied cells is at most $o(1) + (N/s^2)\{\mathbb{P}(|Y| > M) + o(1)\}$, whose limit is $\mathbb{P}(|Y| > M)$, arbitrarily small for M large. No independence is used. □

Remark 5.7. By Proposition 5.6(ii) the S-ersatz tends to one for outputs with unbounded support, even under independence. The approach is slow: for Gaussian and log-normal outputs the simulated values under independence grow from 0.473 and 0.721 at $N = 2^8$ to 0.59 and 0.89 at $N = 2^{16}$ (uniform output: 0.368; output Beta(2, 2): 0.409, against the limit $\int_0^1 e^{-6v(1-v)}\mathrm{d}v = 0.4097$). When X_i and Y are dependent and the output is bounded, the argument of (i) combined with Lemma 4.4 gives the limit $\iint \exp\{-c(u, \tilde{F}(v))\tilde{f}(v)\}\,\mathrm{d}u\,\mathrm{d}v$, where $\tilde{F}$ is the distribution function of the scaled output; it depends on the output marginal as well as on the copula. The S-ersatz is therefore not copula-based in the sense of Plischke and Borgonovo (2019, Table 1): like the correlation ratio, it is invariant under increasing affine transformations of the output but not under other strictly increasing ones. The rank transformation of the first modification removes this dependence (Proposition 3.2(iii) and Theorem 4.1).

6 Grid resolution

Theorem 6.1 (phase transition in α_g)**.** *Let Assumption 2.1 hold, let the sample be drawn by SRS, and let $s = \lceil N^{\alpha_g} \rceil$ with $\alpha_g > 1/4$.*

(i) If $\alpha_g < 1/2$ and $\mathcal{Z} = \{(u, v) \notin K : c_i(u, v) = 0\}$ satisfies $\mathrm{Leb}(\mathcal{Z} \setminus \mathrm{int}\,\mathcal{Z}) = 0$, then $\hat{S}_N^{\circ} \to \mathrm{Leb}(\mathcal{Z})$ and $\hat{S}_N^{\mathrm{adj}} \to \mathrm{Leb}(\mathcal{Z})$ almost surely. In particular both statistics tend to zero for every copula with a positive density, whatever the strength of the dependence (for instance every Gaussian copula with correlation in $(-1, 1)$).

(ii) If $\alpha_g = 1/2$, Theorem 4.1 applies with $\lambda = 1$.

(iii) If $\alpha_g > 1/2$, then $\hat{S}_N^{\mathrm{adj}} \geq 1 - 9N/s^2$ and $\hat{S}_N^{\circ} \geq 1 - N/s^2$, so both statistics tend to one for every copula.

Proof. (iii) At most N cells are occupied, and a cell can be filled only if one of its neighbours is occupied, so at most $8N$ cells are filled. (i) Here $N/s^2 \to \infty$. For $(u,v) \notin K$ with $c_i(u,v) > 0$, the localisation step of Lemma 4.4 gives $\sum_{r \in B_b} \pi_{r,a} \geq (N/s^2)(c_i(u,v) - \eta_N)(1 + o(1)) \to \infty$, so $\mathbb{P}(n_{ab} = 0 \mid \mathcal{V}) \leq \exp(-\sum_r \pi_{r,a}) \to 0$, and the cell is a fortiori not empty after the fill. For $(u,v) \in \operatorname{int} \mathcal{Z}$, for N large the whole neighbourhood D_N lies in $\operatorname{int} \mathcal{Z}$, all $\pi_{r,a'}$ of the 3×3 block vanish, and the cell and its neighbours are empty with probability one, so it stays empty after the fill. Dominated convergence gives the limit of the conditional means. The bounded-difference bounds (4) hold unchanged, and $s^4/N = N^{4\alpha_g - 1}(1 + o(1))$ grows polynomially when $\alpha_g > 1/4$, so they are summable. □

Theorem 6.1 explains why the grid exponent dominates the sensitivity analysis of the method in the main text. At fixed N, $\lambda_N = N/\lceil N^{\alpha_g} \rceil^2$ lies in $\left(N^{-0.2}(1 + N^{-0.6})^{-2},\ N^{0.2}\right]$ when $\alpha_g \in [0.4, 0.6]$, because $N^{\alpha_g} \leq \lceil N^{\alpha_g} \rceil < N^{\alpha_g} + 1$; for $N = 1600$ it ranges from 0.23 ($\alpha_g = 0.6$, $s = 84$) to 4.0 ($\alpha_g = 0.4$, $s = 20$). Over this range the value under independence κ_λ moves from 0.75 to $2 \cdot 10^{-9}$, which shifts all values of the statistic and its screening errors. On the finest grids, $0.0422 < \lambda < \lambda_c = 0.4815$ (for $N = 1600$, when $s \geq 58$, that is, $\alpha_g > \log 57 / \log 1600 = 0.5480\ldots$), κ_λ is no longer a lower bound: dependence can then lower the statistic below its value under independence (Proposition 5.4(iii)). A limit equal to zero for every copula with a positive density holds only in regime (i), not for the default grid. The theorem does not classify $\alpha_g \leq 1/4$, for which the concentration bounds (4) no longer tend to zero. Such grids, with $s \leq \lceil N^{1/4} \rceil$, lie far below the range $\alpha_g \in [0.4, 0.6]$ explored in the main text.

7 Bivariate measures and Sobol' indices

For independent inputs and $Y \in L^2$ with $\operatorname{Var}(Y) > 0$, $S_i = \operatorname{Var}(\mathbb{E}[Y \mid X_i]) / \operatorname{Var}(Y)$ and $T_i = \mathbb{E}[\operatorname{Var}(Y \mid \mathbf{X}_{\sim i})] / \operatorname{Var}(Y)$ (Homma and Saltelli, 1996; Saltelli et al., 2008). By the factor-fixing result of Sobol' (2001), fixing X_i at an independent random value changes the output by a mean squared error equal to $2\,T_i \operatorname{Var}(Y)$. $T_i = 0$ is therefore the operational definition of a non-influential input.

Proposition 7.1 (relations with Sobol' indices)**.** *Let the inputs be independent and $Y \in L^2$ with $\operatorname{Var}(Y) > 0$.*

(i) If $T_i = 0$ then X_i and Y are independent, so $C_i = \Pi$, $\delta_i = 0$, every f-sensitivity index vanishes, $\Phi_\lambda(C_i) = e^{-\lambda}$ and $\Psi_\lambda(C_i) = \kappa_\lambda$.

(ii) The converse fails: for $Y = (X_1 + X_2) \bmod 1$, X_1 and Y are independent, but $T_1 = 1$.

(iii) $S_i > 0$ if and only if $\mathbb{E}[Y \mid X_i]$ is not almost surely constant.

(iv) If $T_i = S_i$ (equivalently, $Y = f^{(1)}(X_i) + f^{(2)}(\mathbf{X}_{\sim i})$ almost surely), then $Y - \mathbb{E}[Y \mid X_i]$ is independent of X_i. Hence, if the conditional distribution of $Y - \mathbb{E}[Y \mid X_i]$ given X_i varies with X_i (for instance if the conditional variance is not constant), then X_i interacts with other inputs, $T_i > S_i$.

(v) The converse of (iv) fails, by the example in (ii).

(vi) A small T_i does not imply a small δ_i: for $Y = \mathbf{1}\{X_2 > 1/2\} + \varepsilon X_1$ with $0 < \varepsilon < 1$, $T_1 = \varepsilon^2/(3 + \varepsilon^2)$, whereas X_1 is a function of Y, so C_1 is singular and $\delta_1 = \Phi_\lambda(C_1) = \Psi_\lambda(C_1) = 1$.

Proof. (i) $T_i = 0$ means $\operatorname{Var}(Y \mid \mathbf{X}_{\sim i}) = 0$ almost surely, so $Y = \mathbb{E}[Y \mid \mathbf{X}_{\sim i}]$ is a function of $\mathbf{X}_{\sim i}$, which is independent of X_i; then V, a randomised function of Y, is independent of X_i too. The remaining claims follow from Propositions 5.1 and 5.4(i). (ii) For every $x \in [0,1]$,

$(x + X_2) \bmod 1$ is uniform, so the conditional law of Y given $X_1 = x$ does not depend on x; and $\mathrm{Var}(Y \mid X_2) = \mathrm{Var}(Y)$, so $T_1 = 1$. (iii) is the definition. (iv) In the ANOVA decomposition $Y = \sum_u f_u(\mathbf{X}_u)$, $T_i - S_i = \sum_{u \ni i, |u| \geq 2} \mathrm{Var}(f_u)/\mathrm{Var}(Y)$, so $T_i = S_i$ if and only if every term with $i \in u$, $|u| \geq 2$, vanishes. Then $\mathbb{E}[Y \mid X_i] = f_\emptyset + f_i(X_i)$ and $Y - \mathbb{E}[Y \mid X_i] = \sum_{u \not\ni i} f_u(\mathbf{X}_u)$, a function of $\mathbf{X}_{\sim i}$. (vi) $\mathrm{Var}(Y) = 1/4 + \varepsilon^2/12$ and $\mathbb{E}[\mathrm{Var}(Y \mid X_2)] = \varepsilon^2/12$. Since $\varepsilon < 1$, $X_1 = (Y - \mathbf{1}\{Y \geq 1\})/\varepsilon$ is a function of Y, so (X_1, V) lies on the graph of a function and C_1 is singular; the last claims follow from Propositions 5.1(i), 5.2 and 5.4(i). □

Proposition 7.1 states precisely what a scatterplot of (X_i, Y), and any statistic computed from it, can reveal. The conditional mean carries the first-order effect. A change of the conditional distribution beyond a location shift, such as a conditional variance that depends on X_i, is the imprint of interactions involving X_i, of any order (for $Y = X_1X_2X_3$ with centred inputs, $\mathbb{E}[Y \mid X_1] = 0$ while $\mathrm{Var}(Y \mid X_1)$ is proportional to X_1^2). Interactions that leave the conditional distribution unchanged, as in (ii), are invisible to every bivariate measure. A bivariate statistic therefore measures the dependence between X_i and Y, which includes but does not exhaust the contributions to T_i.

Theorem 7.2 (single-index Gaussian models)**.** *Let $Z_j = \Phi_{\mathcal{N}}^{-1}(X_j)$, where $\Phi_{\mathcal{N}}$ is the standard normal distribution function, let $a \in \mathbb{R}^k \setminus \{0\}$, $\rho_j = a_j/\|a\|$, $W = \sum_j \rho_j Z_j \sim N(0,1)$, and $Y = g(W)$ with g measurable and $g(W)$ not almost surely constant.*

(i) If $g(W) \in L^2$, with Hermite expansion $g(w) = \sum_{n \geq 0} \gamma_n \mathrm{He}_n(w)/\sqrt{n!}$, then

$$S_i = \frac{\sum_{n \geq 1} \gamma_n^2 \rho_i^{2n}}{\sum_{n \geq 1} \gamma_n^2}, \qquad T_i = \frac{\sum_{n \geq 1} \gamma_n^2 \, [1 - (1 - \rho_i^2)^n]}{\sum_{n \geq 1} \gamma_n^2},$$

both strictly increasing functions of $|\rho_i|$.

(ii) For every convex f with $f(1) = 0$, $|\rho_j| \leq |\rho_i|$ implies $I_f(X_j; Y) \leq I_f(X_i; Y)$. This applies to δ_i, to mutual information and to $\Phi_\lambda(C_i) - e^{-\lambda}$.

(iii) Consequently no f-sensitivity index ranks an input strictly above an input with a larger total-order index: the rankings by T_i, by S_i and by any f-sensitivity index agree up to ties.

Proof. (i) From the generating function $e^{tw - t^2/2} = \sum_n \mathrm{He}_n(w) t^n/n!$ and $\sum_j \rho_j^2 = 1$,

$$e^{tW - t^2/2} = \prod_j e^{t\rho_j Z_j - t^2\rho_j^2/2} \quad \Longrightarrow \quad \frac{\mathrm{He}_n(W)}{\sqrt{n!}} = \sum_{|\alpha| = n} \sqrt{\frac{n!}{\alpha!}} \, \rho^\alpha \prod_j \frac{\mathrm{He}_{\alpha_j}(Z_j)}{\sqrt{\alpha_j!}}$$

(see also Janson, 1997, Chapter 3). The products on the right are orthonormal in L^2, so the variance of the component indexed by α is $\gamma_{|\alpha|}^2 (|\alpha|!/\alpha!) \rho^{2\alpha}$. The total index collects the components with $\alpha_i \geq 1$; by the multinomial theorem $\sum_{|\alpha| = n, \alpha_i \geq 1} (n!/\alpha!) \rho^{2\alpha} = (\sum_j \rho_j^2)^n - (\sum_{j \neq i} \rho_j^2)^n = 1 - (1 - \rho_i^2)^n$. The first-order index collects $\alpha = n\, e_i$. Since $g(W)$ is not constant, some γ_n with $n \geq 1$ is non-zero, which gives strict monotonicity.

(ii) If $\rho_i = 0$ then $\rho_j = 0$ and both informations vanish. Otherwise let $r = \rho_j/\rho_i \in [-1, 1]$ and $Z' = r Z_i + \sqrt{1 - r^2}\, \xi$ with $\xi \sim N(0,1)$ independent of everything else. Then (Z', W) is bivariate normal with unit variances and correlation $r\rho_i = \rho_j$, so (Z', Y) has the law of (Z_j, Y). As Z' is obtained from Z_i through a kernel that does not involve Y, the data-processing inequality (Proposition 5.1(iii)) gives $I_f(Z_j; Y) = I_f(Z'; Y) \leq I_f(Z_i; Y)$, and I_f is invariant under the bijection $\Phi_{\mathcal{N}}$. (iii) combines (i) and (ii). □

Remark 7.3 (the adjusted ersatz and rankings). h_λ is not convex (it is concave where $e^{-\lambda t} \in (p_0, p_1)$), so Ψ_λ is not an f-divergence and Theorem 7.2(ii) does not apply to it. Indeed Ψ_λ violates the data-processing inequality. Let C have, on the 4×4 grid of equal squares, the piecewise

constant density with rows (indexed by u) $(0.1, 0.3, 1.8, 1.8)$, $(0.3, 0.1, 1.8, 1.8)$, $(1.8, 1.8, 0.2, 0.2)$ and $(1.8, 1.8, 0.2, 0.2)$. Averaging the first two rows, which is a Markov kernel acting on X_i only, raises Ψ_1 from 0.3902 to 0.3931, while δ stays at 0.400 and Φ_1 decreases from 0.4930 to 0.4920. Ψ_λ is invariant under strictly increasing transformations of X_i and of Y, and hence copula-based in the sense of Plischke and Borgonovo (2019, Table 1), but it is not one of their niveau 3 measures, which vanish if and only if $C = \Pi$ (their Theorem 13): $\Psi_\lambda(\Pi) = \kappa_\lambda$, and Proposition 5.4(ii) and (iii) give the corresponding property relative to κ_λ. Two facts remain.

(a) By Proposition 5.4(iv), for $\lambda \in [1/2, 1]$, if $\delta_i - \delta_j > \varepsilon_\lambda + \kappa_\lambda$ (0.18711 for $\lambda = 1$) then $\Psi_\lambda(C_i) > \Psi_\lambda(C_j)$. Every law of the density values c (under Π) with mean at most one is the law of a copula: a circulant absolutely continuous part, as in the proof of Proposition 5.4(iii), plus a singular part spread uniformly on the diagonal, that is, a multiple of the copula M of (U, U). The smallest margin that guarantees agreement is therefore the value of a linear programme over two such laws. Solved with density values on a grid of step $5 \cdot 10^{-4}$ on $[0, 3]$, and coarser up to 30, it is 0.1763 for $\lambda = 1$ and 0.1627 for $\lambda = 512/529$. At $\lambda = 1$ it is attained by the copula C_a of Proposition 5.2 against $w\,\Pi + (1 - w)\,M$, where $w = 0.603835$ solves $h_1'(w) = -(1 - \kappa_1)$ and $a^2 = (h_1(w) - \kappa_1)/(1 - \kappa_1) = 0.219831$ (40-digit arithmetic). Their values of δ, a^2 and $1 - w$, differ by 0.176334, and their values of Ψ_1 coincide (0.256563). For a strict reversal, C_a with $a^2 = 0.22$ has $\delta = 0.22$ and $\Psi_1 = 0.2567$, while $0.61\,\Pi + 0.39\,M$ has $\delta = 0.39$ and $\Psi_1 = 0.2507$; at $N = 2^{14}$ the adjusted ersatz gives 0.2505 and 0.2410 (means of 20 replicates). The set of attainable pairs $\big(\delta(C_j) - \delta(C_i),\ \Psi_\lambda(C_i) - \Psi_\lambda(C_j)\big)$ is convex and contains pairs with equal δ and either order of Ψ_λ, so reversals occur for every difference of δ below the sharp margin.

(b) Numerically, $\Psi_1(C_\rho)$ is strictly increasing in $|\rho|$ for Gaussian copulas (240 values of $\rho \in [0, 0.999]$; Section 10). Hence the adjusted ersatz ranks the inputs as T_i does in single-index models with strictly monotone g, for which (X_i, Y) has a Gaussian copula.

Outside single-index models the rankings by T_i and δ_i can differ. For the Ishigami function $T_1 > T_2$ but $\delta_2 > \delta_1$, and the adjusted ersatz follows δ (main text, Section 3.2).

8 Designs other than simple random sampling

Theorem 4.1 is proved for SRS. When Y depends only on other inputs ($T_i = 0$), its conclusion extends to plain Latin hypercube sampling (LHS). For maximin LHS and for randomised quasi-Monte Carlo designs, which neither Theorem 4.1 nor Proposition 8.1 covers, we report simulation evidence only (Table 2). As in the main text, a maximin LHS is the Latin hypercube with the largest minimum Euclidean distance between points among 20 independent random Latin hypercubes (`scipy.stats.qmc.LatinHypercube`), and scrambled Sobol' points are the first N points of `scipy.stats.qmc.Sobol` with its default scrambling, a linear matrix scramble followed by a digital shift.

Proposition 8.1 (Latin hypercube sampling under independence)**.** *Let the column of X_i be a Latin hypercube column: its points are $(k - \omega_k)/N$, $k = 1, \ldots, N$, with $\omega_1, \ldots, \omega_N$ independent $U(0, 1)$ variables, assigned to the runs in a uniformly random order. Let Y be independent of this column, as is the case when Y depends only on other columns of a plain Latin hypercube. If $s \to \infty$ and $N/s^2 \to \lambda \in (0, \infty)$, then $\hat{S}_N^\circ \to e^{-\lambda}$ and $\hat{S}_N^{\mathrm{adj}} \to h_\lambda(1)$ in probability.*

Proof. Condition on the ω_k. Then every row I_a of the grid contains a deterministic set of N_a strata $((k - 1)/N, k/N]$, with $|N_a - N/s| \le 2$. The matching between strata and ranks is a uniformly random bijection, because the column is assigned to the runs in a uniformly random order, independent of the output and hence of its ranks, however ties are broken. Let $\mathcal{C}$ be a finite set of cells and $r_{ab} \ge 0$ integers, $(a, b) \in \mathcal{C}$, with $|r| = \sum r_{ab}$, $r_{a\cdot} = \sum_b r_{ab}$ and $r_{\cdot b} = \sum_a r_{ab}$.

Counting ordered selections of distinct strata in each row and of distinct ranks in each block gives the factorial moments

$$\mathbb{E}\Big[\prod_{(a,b)\in\mathcal{C}}(n_{ab})_{r_{ab}}\Big]=\prod_{a}(N_a)_{r_{a\cdot}}\prod_{b}(|B_b|)_{r_{\cdot b}}\,\frac{(N-|r|)!}{N!},$$

where $(x)_r=x(x-1)\cdots(x-r+1)$. Since N_a and $|B_b|$ are $N/s+O(1)$ uniformly, the right-hand side converges to $\lambda^{|r|}$ uniformly over the positions of the cells, and $\lambda^{|r|}$ is the factorial moment of independent Poisson(λ) variables. As the Poisson law is determined by its moments, the counts of any bounded set of cells converge jointly to independent Poisson(λ) variables. The convergence is uniform in the positions of the cells: otherwise a sequence of positions along which it fails would have factorial moments converging to $\lambda^{|r|}$, hence counts converging to the Poisson limit, a contradiction. Taking for $\mathcal{C}$ the 3×3 block of an interior cell gives $\mathbb{E}\hat{S}_N^{\mathrm{adj}}\to h_\lambda(1)$. Taking two blocks gives $\mathrm{Cov}(\mathbf{1}_c,\mathbf{1}_{c'})\to 0$ uniformly over pairs of interior cells with disjoint blocks, where $\mathbf{1}_c$ indicates that cell c is empty after the fill. Boundary cells and pairs with intersecting blocks are fractions $O(1/s)$ and $O(1/s^2)$ of the cells and of the pairs, so $\mathrm{Var}\,\hat{S}_N^{\mathrm{adj}}\to 0$, and Chebyshev's inequality gives the claim. The argument for $\hat{S}_N^{\circ}$ is the same with single cells. □

The null law of plain LHS does not depend on the model but differs from that of SRS at finite N. In simulations, its mean and its probability of exceeding 0.05 are 0.045 and 0.34 at $N=512$, and 0.044 and 0.12 at $N=4096$, within 0.001 (mean) and 0.02 (probability) of the maximin values of Table 2(b). Proposition 8.1 does not cover maximin LHS, in which the selection by distance couples the columns. In the simulations of Table 2, the values of maximin LHS under independence are below those of SRS at small N and approach κ_{λ_N} as N grows.

Scrambled Sobol' sequences. The points of a scrambled digital net are individually uniform but jointly dependent (Owen, 1997; Dick and Pillichshammer, 2010), so Theorem 4.1 does not apply, and the value of the statistic under independence depends on the model. If Y is a monotone function of a single other input X_l, the rank-transformed pairs (X_i,Y) inherit the two-dimensional net structure of the projection (X_i,X_l) and are far more regular than random points. For the first two coordinates of the sequence at $N=512$ (means of 200 replicates), the unfilled statistic is 0.25 instead of 0.37, and the adjusted ersatz is 0.0004 instead of 0.054; for coordinates 3 and 5 the values are 0.34 and 0.022. If Y depends on many inputs, the values approach those of SRS as N grows (Table 2: for the sum of five inputs, 0.033 against 0.054 at $N=512$, 0.044 against 0.047 at $N=4096$). Conversely, two-dimensional projections of digital sequences with large t-values contain regular patterns of empty cells, which can be mistaken for structure in a scatterplot. Such patterns are a property of the design, not evidence of interactions.

9 Null distribution and the three thresholds

Table 2 reports the distribution of $\hat{S}_N^{\mathrm{adj}}$ for an input independent of the output. Under SRS this distribution depends only on N and s, because the input values are then independent and identically distributed, independently of the output, so panel (a) applies to every model. Under plain LHS it is also free of the model but differs at finite N (Section 8). Under scrambled Sobol' sequences it depends on the model, and we report the case in which Y is the sum of five other inputs. At $N=512$ ($s=23$) the SRS values have mean 0.0536 and standard deviation 0.0155 (rounded to three decimals in Table 2), with 95% and 99% quantiles 0.081 and 0.093, and an independent input exceeds $\theta=0.01$, 0.05 and 0.10 with probabilities 1.000, 0.566 and 0.004.

The three thresholds used in the main text therefore have a precise meaning for the adjusted ersatz under SRS. At $N=512$ the threshold $\theta=0.01$ lies 2.8 standard deviations below the mean under independence, and an input independent of the output exceeded it in 9998 of the 10,000 simulated samples. The threshold $\theta=0.05$ lies 0.23 standard deviations below the mean and was exceeded in 56.6% of them. This share oscillates with $\lambda_N=N/\lceil\sqrt{N}\rceil^2$ (0.35 at $N=1024$, 0.71

at $N = 2048$), because 0.05 lies between $\kappa_1 = 0.047$ and $\kappa_{0.968} = 0.055$. The threshold $\theta = 0.10$ lies 3.0 standard deviations above the mean, beyond the 99% quantile, and was exceeded in 0.37% of the samples. Under the Sobol' design the values are lower and depend on the model: at $N = 512$, for the sum of five inputs, the mean is 0.033 and an independent input exceeds 0.05 with probability 0.095. A threshold calibrated on the statistic's own scale is an upper quantile of its null distribution. Under SRS this quantile depends only on N and can be read from Table 2 or simulated once; for data drawn by SRS or plain LHS, permuting the output gives the null distribution of the design at hand.

(a) Simple random sampling, 10,000 replicates; valid for every model

N	s	λ_N	κ_{λ_N}	mean	sd	$q_{0.95}$	$q_{0.99}$	$\mathbb{P}(\hat{S} > 0.01)$	$\mathbb{P}(\hat{S} > 0.05)$	$\mathbb{P}(\hat{S} > 0.10)$
32	6	0.889	0.079	0.066	0.062	0.194	0.250	0.751	0.572	0.268
64	8	1.000	0.047	0.044	0.039	0.125	0.156	0.810	0.325	0.086
128	12	0.889	0.079	0.073	0.033	0.132	0.160	0.989	0.718	0.198
256	16	1.000	0.047	0.046	0.020	0.082	0.102	0.982	0.412	0.011
512	23	0.968	0.055	0.054	0.015	0.081	0.093	1.000	0.566	0.004
1024	32	1.000	0.047	0.047	0.010	0.064	0.072	1.000	0.347	0.000
2048	46	0.968	0.055	0.054	0.008	0.068	0.074	1.000	0.707	0.000
4096	64	1.000	0.047	0.047	0.005	0.056	0.060	1.000	0.274	0.000

(b) Maximin LHS and scrambled Sobol' sequences, Y the sum of five other inputs

	Maximin LHS				Scrambled Sobol' (10,000 replicates)		
N	replicates	mean	sd	$\mathbb{P}(\hat{S} > 0.05)$	mean	sd	$\mathbb{P}(\hat{S} > 0.05)$
32	2000	0.031	0.041	0.305	0.015	0.028	0.141
64	2000	0.023	0.028	0.131	0.011	0.018	0.032
128	2000	0.053	0.029	0.460	0.030	0.021	0.145
256	2000	0.034	0.018	0.190	0.022	0.014	0.043
512	2000	0.045	0.014	0.334	0.033	0.013	0.095
1024	2000	0.040	0.009	0.144	0.032	0.009	0.034
2048	500	0.049	0.008	0.456	0.044	0.008	0.237
4096	300	0.044	0.006	0.133	0.044	0.007	0.194

Table 2: Adjusted ersatz $\hat{S} = \hat{S}_N^{\mathrm{adj}}$ of an input independent of the output (Monte Carlo simulation). The column $\kappa_{\lambda_N} = h_{\lambda_N}(1)$ is the interior-cell limit; the oscillation with N follows $\lambda_N = N/\lceil\sqrt{N}\rceil^2$. Bootstrap standard errors are at most 0.0006 for the means, 0.009 for the quantiles and 0.005 for the probabilities of panel (a) and of the Sobol' columns, and at most 0.001 for the means and 0.023 for the probabilities of the maximin LHS columns. Maximin LHS: the best of 20 random Latin hypercubes by minimum distance between points (Section 8), with seed $97000N + r$ for replicate r at sample size N.

10 Numerical verification of the limits

Table 3 compares the population functionals with simulated values at $N = 2^{16}$ (SRS, five replicates) for Gaussian copulas. In normal-score coordinates rotated by 45°, $s' = (x + y)/\sqrt{2}$ and $d' = (x - y)/\sqrt{2}$, the Gaussian copula density is $c_\rho = (1 - \rho^2)^{-1/2} \exp\{\rho s'^2/(2(1 + \rho)) - \rho d'^2/(2(1 - \rho))\}$. The functionals were computed by Gauss–Legendre quadrature in these coordinates, with the inner rule restricted to the band in which c_ρ is not negligible, and δ by the formula $\delta = \mathbb{E}_X \,\mathrm{TV}\left(N(\rho X, 1 - \rho^2), N(0, 1)\right)$ with one-dimensional adaptive quadrature. Monte Carlo integration with 10^7 draws agrees to within $2 \cdot 10^{-4}$. A tensor Gauss–Hermite rule is not accurate enough for $\rho \geq 0.99$, where the density concentrates near the diagonal, nor for δ, whose integrand has a kink. The identity of Proposition 3.3 was checked to hold exactly in every

replicate.

ρ	$\Phi_1(C_\rho)$	$\hat{S}_N^\circ$	$\Psi_1(C_\rho)$	$\hat{S}_N^{\text{adj}}$	$\delta(C_\rho)$
0.00	0.3679	0.3676	0.0471	0.0481	0.0000
0.30	0.3842	0.3833	0.0904	0.0905	0.1010
0.60	0.4371	0.4364	0.2143	0.2131	0.2367
0.90	0.5945	0.5929	0.5013	0.4991	0.5149
0.99	0.8180	0.8176	0.7977	0.7967	0.8015

Table 3: Gaussian copulas: limits of Theorem 4.1, simulated statistics at $N = 2^{16}$, and Borgonovo's δ.

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